

PRD_Teams_Module

Teams Module and Team Page

Full Product Requirements Document for Product, UX, Frontend, Backend, QA, Security and DevOps Teams

The Teams Module is the operational collaboration layer connecting users across departments, companies, projects and business functions without changing their primary employment structure.

Global rule: all sections, widgets, features and actions must support Platform Configuration. The initial implementation exposes the full capability set; user-type restrictions will be introduced later.

Status: Confirmed product scope
Version: 1.0
Audience: Full-stack implementation team

1. Document Control

Field	Value
Document	PRD_Teams_Module
Primary pages	Teams Module Page, Team Page
Parent context	Group, Company, Department
Connected modules	Users, Projects, Tasks, Documents, Meetings, Approvals, Calendar, Assets, Clients, Risks, Reports
Access model	Full capability first; restrictions later
Platform Configuration	Mandatory compatibility at module, page, section, widget, feature and action level

2. Product Vision

- Create a universal team structure for permanent, temporary, project, executive, cross-department and cross-company teams.
- Allow a user to belong to multiple teams without changing employment ownership.
- Connect team members to projects, tasks, documents, meetings, assets, clients, risks and KPIs.
- Provide clear workload, skills and capacity visibility.
- Keep task access participant-based rather than automatically exposing tasks to every team member.
- Support any business category through configurable team types and roles.

The Team Page is an operational workspace, not only a member directory.

3. Scope and Boundaries

- In scope: Teams Module, Team Page, membership, roles, assignments, workload, skills, assets, risks, KPIs, reports, settings and activity.
- In scope: responsive desktop, tablet and mobile behavior.
- In scope: data model, APIs, validation, audit, performance and testing.
- Out of scope: final permissions matrix, third-party integrations, AI-generated performance judgments and full payroll logic.

4. Information Architecture

- Teams Module Page
- Team Page
- Create Team
- Edit Team
- Team Members
- Team Roles
- Projects
- Tasks
- Documents
- Meetings
- Approvals
- Calendar
- Workload and Capacity
- Skills Matrix

- Assets
- Clients
- KPIs
- Risks
- Reports
- Activity
- Settings

5. Team Types

- Permanent
- Temporary
- Project team
- Cross-department
- Cross-company
- Executive
- Sales
- Design
- Construction
- Site
- Procurement
- Support
- Custom

Team types are configurable and must not be hard-coded.

6. Team Roles

- Team leader
- Deputy leader
- Coordinator
- Reviewer
- Contributor
- Specialist
- External member
- Observer
- Custom role

Team roles describe responsibility within the team and do not replace the user's employment position.

7. Teams Module Header

- Page title
- Create Team
- Global Search
- Filters
- Grid/List switch

- Saved views
- Export
- Platform Configuration status

8. Teams Module Summary Cards

- Total teams
- Active teams
- Temporary teams
- Cross-company teams
- Total members
- Team leaders
- Active projects
- Open tasks
- Overdue tasks
- Average workload

Clicking a summary card applies the corresponding filter.

9. Team Cards and Rows

- Team name
- Team code
- Team type
- Department
- Company
- Team leader
- Member count
- Active projects
- Open tasks
- Workload
- Availability
- Health indicator
- Status
- Personal Star

Clicking a team opens the Team Page.

10. Teams Module Filters

- Group
- Company
- Department
- Team type
- Team leader
- Status
- Member count
- Active projects

- Workload
- Availability
- Health
- Recently updated
- Starred by me

11. Teams Module Actions

- Create Team
- Edit Team
- Duplicate Team
- Archive
- Restore
- Add members
- Remove members
- Assign project
- Export
- Open Team Page

12. Team Header

- Team name
- Icon/photo
- Team code
- Team type
- Company
- Department
- Parent team
- Team leader
- Deputy leader
- Status
- Start date
- End date
- Description

13. Team Header Actions

- Edit Team
- Add Member
- Assign Project
- Create Task
- Upload Document
- Schedule Meeting
- Add Asset
- Add Risk
- Open Settings

- Archive Team

14. Overview Dashboard

- Members
- Projects
- Tasks
- Documents
- Meetings
- Approvals
- Workload
- Availability
- Assets
- Risks
- KPIs
- Recent activity
- Team health indicators

Every widget is individually configurable through Platform Configuration.

15. Members

- Name
- Position
- Company
- Department
- Team role
- Membership type
- Allocation percentage
- Start/end dates
- Skills
- Certifications
- Availability
- Workload
- Current tasks
- Current projects
- Status

Clicking a member opens User Profile.

16. Membership Rules

- A user may belong to multiple teams.
- Membership may be permanent or time-limited.
- Cross-company membership is allowed when business rules permit.
- Membership does not change the user's employer, department or employment record.
- Each membership stores allocation, role and effective dates.
- Removing a member does not delete historical task, document or activity records.

17. Projects

- Assigned projects
- Team role
- Project manager
- Team allocation
- Start/end dates
- Progress
- Milestones
- Tasks
- Documents
- Risks
- Status

Project allocation and team allocation are separate concepts and must not overwrite each other.

18. Tasks

- List, Board, Calendar and Timeline shortcuts
- Task title
- Participants
- Responsible user
- Project
- Priority
- Status
- Due date
- Progress
- Overdue warning

Tasks remain participant-based. Team membership alone does not grant task access.

19. Documents

- Team Policies
- Templates
- Working Documents
- Reports
- Meeting Minutes
- Project Documents
- Deliverables
- Internal Notes

Uses the central Documents Module; no separate file store.

20. Meetings

- Upcoming
- Previous

- Recurring
- Project meetings
- Internal team meetings
- Minutes
- Decisions
- Action items
- Linked tasks
- Attachments

21. Approvals

- Pending
- Awaiting team member
- Approved
- Rejected
- Returned
- Overdue
- Linked documents/tasks

22. Calendar

- Meetings
- Task deadlines
- Project milestones
- Member leave
- Deliveries
- Reviews
- Training
- Team events

23. Workload and Capacity

- Workload by member
- Allocation by project
- Available capacity
- Over-allocation
- Upcoming availability
- Unassigned work
- Heatmap
- Capacity warnings

Workload calculations must show their source data and should not be treated as employee-performance judgments.

24. Skills Matrix

- Skills by member
- Proficiency

- Certifications
- Skill gaps
- Required skills
- Backup resources
- Training needs

25. Assets

- Devices
- Vehicles
- Tools
- Machinery
- Software licenses
- Shared equipment
- Assigned member
- Assigned project
- Condition
- Maintenance status

26. Clients

- Assigned clients
- Team role
- Active opportunities
- Client tasks
- Client meetings
- Documents
- Sales value
- Issues

This section is optional and can be disabled for non-client-facing teams.

27. KPIs

- Task completion
- Overdue rate
- Project delivery
- Team utilization
- Capacity
- Response time
- Quality issues
- Approval time
- Client satisfaction
- Budget usage
- Skill coverage

KPIs are configurable and must show formula/source.

28. Risks and Issues

- Title
- Category
- Probability
- Impact
- Priority
- Owner
- Mitigation
- Due date
- Status
- Linked project/task/document

29. Reports

- Weekly
- Monthly
- Quarterly
- Workload
- Tasks
- Projects
- Members
- Skills
- KPIs
- Risks
- Assets

30. Activity Timeline

- Team created
- Leader changed
- Member added/removed
- Project assigned
- Task created/completed
- Document uploaded
- Meeting held
- Approval completed
- Asset assigned
- Risk opened/closed
- Team archived/restored

Every event stores acting user, timestamp, source module and linked record.

31. Team Settings

- Name and icon
- Team type

- Company
- Department
- Parent team
- Leader
- Working hours
- Calendar
- Default folders
- Notification defaults
- Team roles
- Status
- Archive/restore

32. Platform Configuration Compatibility

- Module enable/disable
- Page enable/disable
- Section enable/disable
- Widget enable/disable
- Feature enable/disable
- Action enable/disable
- Report enable/disable
- Automation enable/disable
- Team type configuration
- Team role configuration

Disabled functionality must disappear from navigation, search, APIs and layouts without dead links or blank spaces.

33. Responsive UX

- Desktop: wide header, sticky section navigation, multi-column overview, tables and preview drawers.
- Tablet: collapsible navigation, two-column overview, drawers for details.
- Mobile: compact sticky header, single-column cards, scrollable tabs, bottom action sheet, card-based lists.
- Mobile must retain all core functionality.

34. Premium Motion and Micro-interactions

- Team card hover: 120-180 ms lift, shadow and border tint.
- Member avatar stack: subtle reveal and tooltip.
- Tab switch: underline glide and content crossfade.
- Workload heatmap: smooth interpolation.
- Skills matrix cells: hover reveal with details.
- Project assignment: optimistic insertion and confirmation.
- Preview drawers: 220-280 ms slide and fade.
- Loading: skeletons matching final geometry.
- Respect reduced-motion settings.

35. Empty, Loading and Error States

- No teams
- No members
- No projects
- No tasks
- No documents
- No meetings
- No skills data
- No assets
- Module disabled
- Source module unavailable
- Partial widget failure
- Archived team

36. Data Model

Entity	Purpose	Key fields
Team	Primary team record	id, tenant_id, company_id, department_id, parent_team_id, name, code, type, leader_id, deputy_id, start_at, end_at, status
TeamMembership	User-team relation	team_id, user_id, role_id, membership_type, allocation_pct, start_at, end_at, status
TeamRole	Configurable role	id, tenant_id, name, description, status
TeamType	Configurable type	id, tenant_id, name, category, status
TeamProjectAssignment	Team-project relation	team_id, project_id, role, allocation_pct, start_at, end_at
TeamAssetAssignment	Asset relation	team_id, asset_id, assigned_user_id, project_id, start_at, end_at
TeamSkillRequirement	Required skill	team_id, skill_id, required_level, required_count
TeamKPI	KPI definition/value	team_id, metric_id, period, target, actual, status
TeamRisk	Risk relation	team_id, risk_id
TeamStar	Private star	team_id, user_id, created_at
TeamActivity	Activity event	team_id, actor_id, module, entity_type, entity_id, event_type, occurred_at

37. Data Ownership Rules

- User Profile/HR owns user identity and employment.
- Projects owns project records.
- Tasks owns tasks and participant visibility.
- Documents owns document records and revisions.
- Assets owns asset records.
- The Teams Module owns team identity, membership, roles, types and assignments.
- Summary caches must be reproducible from source records.
- Team removal must not delete historical source data.

38. API Requirements

- GET /teams
- POST /teams

- GET /teams/{id}
- PATCH /teams/{id}
- POST /teams/{id}/archive
- POST /teams/{id}/restore
- GET /teams/{id}/overview
- GET /teams/{id}/members
- POST /teams/{id}/members
- PATCH /teams/{id}/members/{userId}
- DELETE /teams/{id}/members/{userId}
- GET /teams/{id}/projects
- POST /teams/{id}/projects
- GET /teams/{id}/tasks
- GET /teams/{id}/documents
- GET /teams/{id}/meetings
- GET /teams/{id}/approvals
- GET /teams/{id}/calendar
- GET /teams/{id}/workload
- GET /teams/{id}/skills
- GET /teams/{id}/assets
- GET /teams/{id}/clients
- GET /teams/{id}/kpis
- GET /teams/{id}/risks
- GET /teams/{id}/activity
- GET /team-types
- POST /team-types
- GET /team-roles
- POST /team-roles

Aggregated endpoints should return summaries and deep links rather than full copies of every connected record.

39. Frontend Architecture

- TeamsModulePage
- TeamsSummaryCards
- TeamsFilterBar
- TeamsGrid
- TeamsTable
- TeamCard
- TeamPage
- TeamHeader
- TeamOverviewDashboard
- TeamMembersPanel
- TeamProjectsPanel
- TeamTasksPanel
- TeamDocumentsPanel

- TeamMeetingsPanel
- TeamApprovalsPanel
- TeamCalendarPanel
- TeamWorkloadPanel
- TeamSkillsMatrix
- TeamAssetsPanel
- TeamClientsPanel
- TeamKPIPanel
- TeamRisksPanel
- TeamReportsPanel
- TeamActivityTimeline
- TeamSettingsPage

40. Backend Services and Jobs

- Team CRUD service
- Membership service
- Role/type configuration service
- Project assignment service
- Workload aggregation service
- Skills matrix service
- Asset assignment service
- KPI calculation service
- Risk aggregation service
- Activity feed service
- Search indexing service
- Configuration resolver
- Cache invalidation service

Background jobs must be idempotent and tenant-aware.

41. Validation Rules

- Team name required
- Team code unique in configured scope
- Leader must be an active user
- Start date cannot follow end date
- Parent-team hierarchy cannot be circular
- Membership date ranges must be valid
- Allocation percentage must be between 0 and 100
- Cross-company membership must respect tenant boundaries
- Duplicate active membership must be prevented
- Archive requires impact confirmation
- Archived teams cannot receive new assignments

42. Security and Permission Readiness

- Tenant isolation
- Company and department scope validation
- Membership data audit
- No automatic access expansion to linked tasks/documents
- Section-level visibility readiness
- Field-level visibility readiness
- Action-level permission readiness
- Platform Configuration and permissions evaluated separately

43. Performance and Scalability

- Load header and core summary first
- Lazy-load heavy sections
- Cursor pagination for members/tasks/activity
- Virtualize long member and project lists
- Avoid N+1 aggregation queries
- Precompute workload and selected KPI summaries
- Support thousands of teams and members
- Target initial interactive load below 2 seconds under normal enterprise conditions

44. Testing Strategy

Unit tests

- Team validation
- Membership date logic
- Allocation validation
- Hierarchy cycle prevention
- Role/type configuration
- Archive/restore
- Platform Configuration resolution

Integration tests

- Users integration
- Projects integration
- Tasks privacy
- Documents linkage
- Assets linkage
- Workload aggregation
- Activity feed

End-to-end tests

- Create team
- Add member
- Assign project
- Create task
- Upload document
- View workload

- Add skill requirement
- Assign asset
- Disable section through Platform Configuration
- Archive and restore
- Mobile behavior

Visual regression

- Desktop Teams Module
- Desktop Team Page
- Tablet
- Mobile
- Long team name
- Large membership list
- Disabled-section reflow
- Reduced-motion mode

45. Acceptance Criteria

- AC-01 - Users can create, edit, archive and restore teams.
- AC-02 - Users can belong to multiple teams.
- AC-03 - Membership does not change employment ownership.
- AC-04 - Cross-department and cross-company teams are supported.
- AC-05 - Team types and roles are configurable.
- AC-06 - Team Page connects all defined modules.
- AC-07 - Team tasks preserve participant-based visibility.
- AC-08 - Project and team allocation remain separate.
- AC-09 - Skills and workload summaries are available.
- AC-10 - Assets, risks and KPIs are linkable.
- AC-11 - Activity is auditable.
- AC-12 - Desktop, tablet and mobile support core actions.
- AC-13 - Premium animations are implemented.
- AC-14 - Reduced-motion preferences are respected.
- AC-15 - Every section is Platform Configuration compatible.
- AC-16 - Disabled features leave no dead links or blank spaces.
- AC-17 - Large teams remain performant.

46. Page and Flow Registry

- Teams Module Page
- Team Page
- Create Team
- Edit Team
- Duplicate Team
- Archive Team
- Restore Team
- Add Team Member

- Remove Team Member
- Team Role Setup
- Team Type Setup
- Assign Team to Project
- Team Members
- Team Projects
- Team Tasks
- Team Documents
- Team Meetings
- Team Approvals
- Team Calendar
- Team Workload
- Team Skills Matrix
- Team Assets
- Team Clients
- Team KPIs
- Team Risks
- Team Reports
- Team Activity Timeline
- Team Settings

47. Implementation Phases

- Phase 1: Team entity, types, roles, membership and Platform Configuration.
- Phase 2: Teams Module Page, Team Page and responsive UI.
- Phase 3: Projects, tasks, documents, meetings and approvals.
- Phase 4: Workload, skills, assets, clients, KPIs and risks.
- Phase 5: Reporting, performance optimization, audit and test hardening.

48. Final Architectural Rule

The Teams Module must remain a universal operational layer that supports multiple team models, preserves employment and source-module ownership, and remains fully compatible with Platform Configuration and future permissions.